Final Project: Pick a Project

Computer Science - Week 10

Oct 3, 2026 - Version 0.0.2

Please make sure to your UD **email** AND **name** is below. Unless your UD emails are included in this table, then you will not earn any points for this assignment when it is graded! **Every person should submit their own individual worksheet**.

Talk about each of the questions with a friend or our LLM (or both), but **make sure the words you enter below are yours** (with the obvious exceptions notwithstanding). Brainstorming is best done in full sentences, and the main goal here is brainstorming. It's worth reviewing the information in this series of pages in the Drafter documentation to gain a better understanding of the process.

Role	Name	Email
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Part 1: The Goal & Prior Implementations

1. **Identify the Goal:** What specific goal are you trying to address? Think about your passions, hobbies, and academic interests. The latter are an excellent candidate to focus on. Some majors have suggestions listed below: these were provided by faculty in those departments. Other majors could consider a focus tied to a big final assignment in one of their recent courses in their major.
    - **Biomedical Engineering** majors may consider one of the following:
        - If you have started (or completed) the Cell/Tissue Design final project in BMEG100, design an organoid, which involves combining a cell type, biomaterial, and testing method. Have the website include a description and diagram of your design concept, a testing method, a needs statement, and an impact statement relevant to their organoid. Also include a link to a paper that supports your idea and a paragraph describing how that paper supports your design concept.

        - Otherwise, create a website related to your BMEG100 Lab Report 1, *Cytoskeletal and Nuclear Staining Lab Report Guidelines*, linked here (University of Delaware account required to access).
    - **Chemistry** and **Chemical Engineering** majors may consider creating an automated chemistry assistant that takes balanced reaction equations, starting reactant masses, and a harvested product amount to determine the limiting reactant, theoretical yield, and percent yield based on principles from Chemistry 2e (OpenStax) — Chapter 4: Chemical Reactions and Stoichiometry. To avoid manually compiling atomic mass tables or writing custom string tokenizers, use the ChemPy library to parse molecular formulas into exact molecular weights and verify reaction stoichiometry against compound data from the NIH PubChem Database. The program outputs an organized breakdown of limiting reagent status, product yields, and remaining masses of unreacted species. Also use the heat of the reaction to determine if it is exothermic or endothermic, and compute the heat flow per mole of the product.
    - Majors in the **Electric & Computer Engineering** department may consider a steganography project that hides messages in images, and then retrieves those messages from uploaded images.
    - **Mechanical Engineering** majors may consider a project that somehow uses both the NumPy and SciPy Python libraries.

I want to create a mechanical engineering focused website that can compare common materials by density, strength, and cost when the user inputs a certain amount of the material. This can help users figure out costs for certain materials for a project, possibly influencing what choices are made.

2. **Prior Implementations:** Search the internet. What existing tools or websites currently attempt to address it?

Matweb is a website tool that allows users to look up and compare materials. My project would be much simpler, with an interface that is easier to use and would display less information.

3. **Your Implementation:** Briefly describe the legitimate purpose or functionality of your site. How would you want this web application to be used?

My website would let users access a basic list of common engineering materials and compare their costs and properties based on a given size. With calculations I could return

the mass and price of a portion of said material that a user wishes to use for a manufacturing part. I would hope to make this a smaller scale simpler implementation compared to Matweb.

Part 2: The Input-Process-Output (IPO) Model

Think about the fundamental flow of your program.

1. **The Core Goal:** Every program has three main phases: taking input, processing data, and presenting output. In one sentence, what is the ultimate goal your program will accomplish for the user? Be specific and use CS1 concepts.

The ultimate goal of my program would be to take user inputs in the from of the name of a material, process the data in the form of comparing costs, loops, and conditional statements to compare properties of materials and calculate mass. The user would be returned the calculations and some string values for properties.

2. Input: What specific information will the user need to provide to your program? (e.g., a city name, a numerical budget, a list of hobbies).

The user would need to input the name of a material that I have information for in my program. They would also add the volume of desired material.

3. Processing: At a high level, how will the computer manipulate or evaluate that data? (e.g., calculate the average, filter out options that are too expensive, compare the input against a database).

The program would have stored material properties using dataclasses assigned to names of common materials. Each data class could contain density, cost, and certain properties. Mass can be processed by multiplying the volume given by the density.

4. Output: What will the user get back? (e.g., a recommendation, a calculated safety factor, a formatted schedule).

The website would display desired material properties, calculated mass of the material, price by desired volume, and a comparison to other materials that may also meet the requirements of the used. The best option could then be identified to make a helpful suggestion for what to use.

Part 3: Data Abstraction

Identify the "nouns" of your problem.

1. **Representing Reality:** Abstraction is the process of representing the real world inside a computer by ignoring unnecessary details. What are the core "things" (nouns) your program needs to track? (e.g., *If building a recipe app, the nouns are "Recipe" and "Ingredient"*).

The main things that I will need to track in my program are the names of materials, their basic properties, and the considerations that a user wants to search for. If a user wants cheap materials, I need some materials to be qualified as “cheap”

2. **Identifying Properties:** For the most important "thing" you listed above, what 3 to 4 specific details do you actually need to solve your problem, and what details will you abstract away?
    - *Details I need:* (e.g., *Ingredient name, weight, cost*)
    - *Details I will ignore:* (e.g., *Ingredient color, brand, expiration date*)

Details i need: Name, density, cost, strength *Ignore: Color, chemical composition, source*

Part 4: Challenges, AI Strategy, & Goals

1. **Difficulties:** What parts of this project seem the most difficult or technically complex?

I think the most difficult portion of this project will be organizing my materials into categories that can be referenced for comparison. If the user wants to search for the cheapest materials for a job, I would like to provide multiple cheap options that are comparable over a few different qualities.

2. **Coding Strategy:** If you will be using an LLM to help code, how will you ensure you understand the generated code and avoid "hallucinations" or "shallow learning" (copying without understanding)? If you won't be using an LLM to help code, this project might take about 4 hours a week to complete, so list each of the days and times in which you work on the project.

If I end up using an LLM in my code I hope to only use it to guide me in the coding process, not generating mass sections of my code. If I am seriously struggling with a section I would likely use an LLM for fixing errors and giving me a boost in order to figure out my next steps.

3. **Goals:** What is your ultimate goal for this project? (e.g., *Impress the graders with a creative UI, build something I can use in my other classes, learn Drafter deeply*).

My main goal is to just create a functional website that can meet all project requirements while hopefully being interesting to me and my peers as it is related to my major. If successful I may use this in the future as I encounter design challenges in mechanical engineering.

Part 5: Research & Brainstorming Conversation

Have a conversation with a Generative AI tool (like ChatGPT or Claude) OR a friend about your project idea. Do not ask for code. Instead, use the AI/friend as a sounding board to research your problem.

1. **The Initial Prompt/Question:** Ask the AI or your friend to help you brainstorm. (e.g., *"I want to build a simple web tool to help students track their study hours. What are 3 common problems students face when tracking their time, and what are 3 simple features a basic tool should have?"*). Paste your prompt below:

Replace this text with your answer.

2. **The Response:** Briefly summarize the best ideas the AI or your friend suggested. What existing solutions or prior apps did they mention?

Replace this text with your answer.

3. **The Pushback:** You must reply to the AI/friend to narrow the scope when it's over-ambitious. Pick the most complex feature they suggested and explain why it is too complicated for an introductory programming project. Ask them for a simpler alternative. *Paste your response here:*

Replace this text with your answer.

4. **Reflection:** Based on this conversation, how has your project idea changed or become more focused?

Replace this text with your answer.

Part 6: Resource Planning

1. What resources will you use if you get stuck?

If I get stuck I plan on looking back at all the sample questions we have successfully completed in order to relearn what finished groups of code should look like that I can implement into my project. Additionally if I get stuck during practicum I can ask one of our TA's for help with anything, and I can also use the LLM within reason to gain information and work through issues.

2. If your LLM provides code that causes a Drafter or Bakery error, or if you run into a problem during deployment, what is your step-by-step debugging plan to resolve it?

If my LLM provides code that causes an error, I would read the error message in order to try and understand what line of code is the issue, and what the issue is. When I identify the area that is faulty I can try to fix it on my own, depending on if it is a simple syntax fix or not. If there is a larger problem with my code i can possibly use the LLM to help debug my code, or work with a classmate or TA.

Part 7: LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

I did not use the LLM for this assignment.

Part 8: Submit

In Google docs, with this document open, please click on *File* > *Download* > *Microsoft Word (.docx)*, and make sure you put that download somewhere you can find it again. Then, upload it to the Canvas assignment about selecting or brainstorming your final project.

When you're done with this assignment, review this page of the Drafter documentation to better prepare for the next step.